the battery can accept charge from the charger and also provide charge to the application. So, if the charger is connected, while the battery is still supplying to the load, it is obvious that only a part of the charge goes to the battery, while the rest is used by the application. Thus, in this case, the battery is not in a discharging mode.

It may also happen that, if a heavily discharged battery is connected in the above arrangement, all the charge is first absorbed by the battery. Here the application suffers until such time that the battery voltage rises to a certain level and the charging supply has excess charge available to power the application. Users must have seen this happening when their mobile phones reach very low levels, and a charger is connected. For some time, until the charging indicator shows a small percentages of battery capacity (typically 2% or more), the mobile phone does not switch on completely.

When the charging voltage is removed, and the load is still connected, the battery enters the discharge mode. So, even though technically the battery may allow both charging and discharging path to be connected, the net battery current at any point can only be either positive (charging) or negative (discharging), but not both.

As mentioned earlier, the BMS also controls the charging and discharging process by ascertaining the running charge capacity of the battery by using a simple technique called coulomb counting. Coulomb counting, at its simplest, is the count of the amount of charge that is present in the battery and is a way to track the state of charge of a battery.

Mathematically, coulomb counting refers to the integral of the current over time. In discrete terms, coulomb counting is a Sigma function, that is:

Σ(current_in_amps × hours_in_operation)

The BMS keeps a running record of the current levels over a given time period, and thereby tries to estimate the relative charge capacity of the battery. The important word to note here is 'relative,' because coulomb counting gives an account of the change in the charge capacity of the battery. Therefore, unless the BMS is calibrated at either ends of the charge/discharge curve, it is not possible for the BMS to estimate the actual state of charge or depth of discharge of the battery.

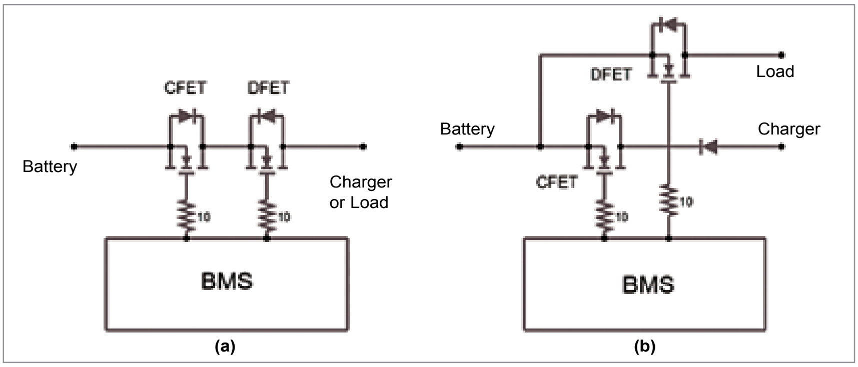

Fig. 3: Cut-off FET schematic illustrations for (a) single connection for the load and charger and (b) two-terminal connection that allows for charging and discharging simultaneously

Barring a few important limitations, coulomb counting can be an effective technique to track the SoC of the battery. The limitations are: (a) the leakage current or self-discharge of the battery, which will not be registered by the BMS, and (b) the offsets in current measurements.

As already mentioned, BMSes solve these issues by using high granularity analogue front-ends and calibrating themselves at the end and start of the charge and discharge curve. Using a BMS of the same current capacity as the battery also helps in this process of tracking the SoC. Thus, at low charge capacities, the BMS can initiate a battery disconnect, even though the discharge current is less than the discharge current limit.

To know the charge and discharge current separately, the BMS may use a single current-sense element or two different current-sense elements, and this is left to the BMS designer. The BMS designer must make a trade-off here between cost and convenience of measurement. The BMS is said to be operating in the charge controller mode in this case.

Maintaining cell voltage

A battery is typically made from series/parallel combination of cells. Parallel connected cells form a local closed circuit and stay at the same voltage. Any difference in their voltage gets distributed based on the Ohms Law. However, series connected cells are not intrinsically in a closed circuit, and hence they can assume different maximum and minimum voltages. This means that a BMS needs to look at individual series cells (or packs of parallel cells) and ensure that the cells stay within their specified voltage range.

But why will cells move to different voltage levels? Here comes the law of probability of microscopic differences between cells, which compound to make one cell weak and another cell strong. A weak cell is one that gains voltage faster than others and loses voltage faster than others. Fig. 4 shows a condition where individual cell differences combine to create widely differing voltages on individual cells, whilst the overall battery voltage remains the same.

Fig. 4 shows various possibilities of charge distribution within a 12V LiFePO4 battery pack. As shown in Fig. 4(c), a weak cell (C-3) causes the battery pack to reach full charged voltage before all the cells have reached their full voltage. This causes